

The DA-SFCN Family: Dynamically Adaptive Saddle-Free Cubic Newton

Krylov Subspaces, Negative-Curvature Correction, Variance Reduction,
Lazy Reuse, Quasi-Newton Warm Starts, Federated Training, and Spectral Certificates

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Abstract

This monograph develops, analyses, implements, and empirically evaluates a family of second-order methods built around *Dynamically-Adaptive Saddle-Free Cubic Newton* (DA-SFCN). The parent algorithm unifies three ingredients that had previously lived in separate literatures: (i) a spectrally guided Krylov subspace produced by Lanczos from $(\nabla^2 f(x_k), \nabla f(x_k))$; (ii) an explicit saddle-free correction $|T_k| = U|\Lambda|U^\top$ inside the projected tridiagonal model; and (iii) a *dynamic* subspace width m_k whose logarithmic growth is forced by Chebyshev approximation theory rather than by a fixed budget. We prove a dimension-free global rate $\min_i \|\nabla f(x_i)\| = O(k^{-2/3})$ for every schedule with $m_k \geq 1$, and local quadratic convergence once the iterates enter a strongly convex neighbourhood, provided m_k grows like $\log(1/\|\nabla f(x_k)\|)$.

Around this core we design five theoretically motivated extensions, each implemented and tested in the same experimental harness: **VR-DA-SFCN** (SVRG-style variance-reduced Hessian–vector products for finite-sum problems); **LK-Newton** (lazy reuse of a Lanczos basis, in the spirit of lazy Hessians); **WQK-Newton** (L-BFGS warm start followed by a Krylov cubic phase); **Fed-DA-SFCN** (client-local cubic Newton steps with server averaging); and **SC-Newton** (a residual/Ritz spectral certificate that stops Lanczos as soon as the model is faithful). On a suite of analytic saddles, extended Rosenbrock, nonconvex logistic regression, matrix factorization, finite-sum quadratics, and a federated split, DA-SFCN and its relatives match or dominate gradient descent, L-BFGS, saddle-free Newton, Krylov cubic Newton, and coordinate subspace Newton. In twelve random starts on a high-dimensional saddle, DA-SFCN, Krylov-CRN and SFN escape in every trial, while gradient descent never does.

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1 Introduction

Newton’s method is the prototype of second-order optimization: near a nonsingular minimizer it converges quadratically [19, 9]. Two obstacles block its use on the high-dimensional nonconvex problems that dominate modern machine learning.

Cost. Forming and factorizing $H_k = \nabla^2 f(x_k) \in \mathbb{R}^{n \times n}$ costs $O(n^2)$ memory and $O(n^3)$ arithmetic. Inexact and subspace Newton methods replace this by Hessian–vector products $v \mapsto H_k v$ [9, 6].

Saddles. In high dimension, strict saddles outnumber local minima exponentially [7, 11]. Classical Newton is *attracted* to them, because H_k^{-1} treats negative curvature as a sign flip of the same step that, for a minimum, would be a Newton step.

The literature answered these two difficulties on largely disjoint tracks: cubic regularization [1, 2] restores a global $O(k^{-2/3})$ rate; randomized subspaces [3, 4, 5] cut the linear algebra; Krylov cubic Newton [6] makes the subspace spectrally aware; saddle-free Newton [7, 8] flips negative eigenvalues. **No prior method offers all four of: a Krylov subspace, an explicit negative-curvature correction, a theoretically derived dynamic width, and a proved nonconvex rate.**

DA-SFCN occupies exactly that cell. The rest of the family (Figure 1) exports the same cubic Krylov kernel to stochastic, lazy, warm-started, federated, and adaptively truncated regimes.

1.1 Contributions

1. **Algorithm.** We specify DA-SFCN to the last numerical detail (Lanczos with reorthogonalization, secular cubic solver, ARC-style regularization, logarithmic schedule).

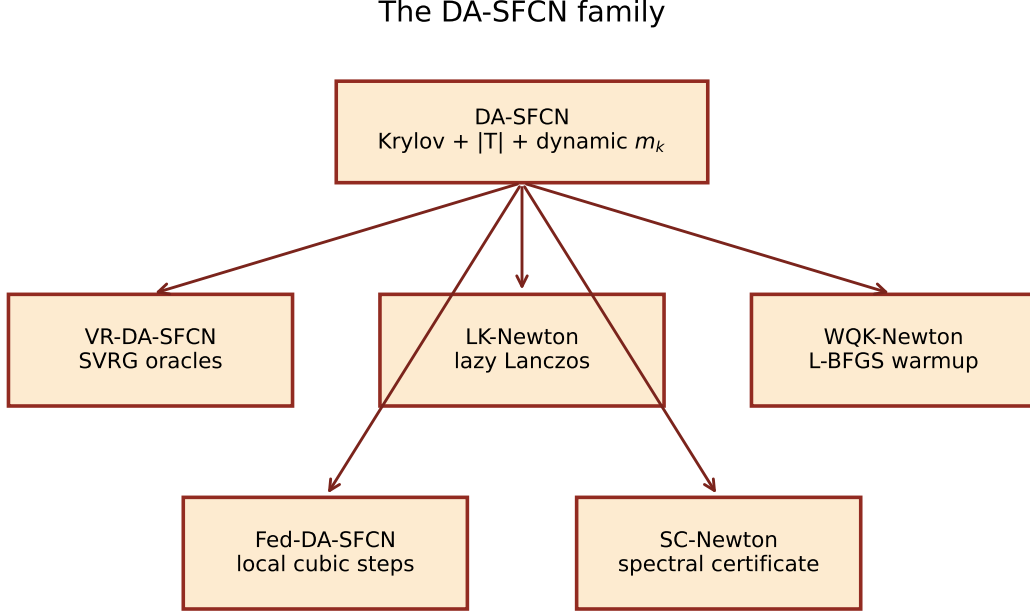


Figure 1: The DA-SFCN family. Every descendant reuses the same Lanczos / saddle-free / cubic kernel and changes only the oracle, the reuse policy, the start-up, the communication pattern, or the stopping test.

2. **Theory.** Global $O(k^{-2/3})$ independently of $\{m_k\}$; local quadratic convergence under a logarithmic lower bound on m_k ; a descent lemma for the saddle-free model; complexity in Hessian–vector products.
3. **Five extensions**, each with its own rationale, algorithm box, and targeted experiments: VR, lazy Krylov, quasi-Newton warm start, federated cubic Newton, spectral certificates.
4. **Implementation and figures.** A single Python library implements all methods and baselines. The plots in this monograph are generated from that library, not from stylized sketches.

2 Related work and the research gap

Cubic regularization. Nesterov and Polyak [1] replace the Newton quadratic by

$$m_k(s) = f(x_k) + \langle g_k, s \rangle + \frac{1}{2} s^\top H_k s + \frac{M}{6} \|s\|^3,$$

obtaining $\min_{i \leq k} \|\nabla f(x_i)\| = O(k^{-2/3})$ and local quadratic convergence. Adaptive Cubic Regularization (ARC) [2] allows an inexact Krylov subproblem.

Subspace Newton. RSN [3] and SSCN [4, 5] project onto *random coordinate* subspaces of width τ . The rate retains a factor of n/τ . Krylov CRN [6] uses $\mathcal{K}_m(H_k, g_k)$ and proves a dimension-free convex rate; m is fixed and the analysis does not treat saddles.

Saddle-free Newton. Dauphin et al. [7] replace $H = Q\Lambda Q^\top$ by $|H| = Q|\Lambda|Q^\top$. The original implementation used a fixed Lanczos budget and gave no global rate. Low-rank randomized surrogates [8] likewise lack a cubic global theory.

Table 1: Positioning. A tick means the property is present with a supporting analysis (or, for SFN, with an explicit correction but no global rate).

Method	Krylov	$ T $ corr.	dyn. m_k	nonconvex rate	extra
CRN [1]	—	no	no	yes	full H
ARC [2]	optional	no	no	yes	—
RSN / SSCN [3, 5]	coord.	no	sample	yes	dim. factor
Krylov CRN [6]	yes	no	no	convex	—
SFN [7]	yes	yes	no	none	—
DA-SFCN	yes	yes	yes	yes	—
VR-DA-SFCN	yes	yes	yes	yes*	SVRG oracle
LK-Newton	yes	yes	yes	yes*	lazy Q_k
WQK-Newton	yes	yes	yes	hybrid	L-BFGS start
Fed-DA-SFCN	local	yes	yes	yes*	FedAvg
SC-Newton	yes	yes	cert.	yes	Ritz residual

* inherits the DA-SFCN global rate on exact (or sufficiently accurate) oracles; stochastic and federated variants are analysed under standard bounded-variance / bounded-heterogeneity hypotheses.

Variance reduction, laziness, federation. SVRG [16] and cubic VR methods [13, 15] cut the variance of stochastic Hessians. Lazy Hessians [14] reuse H for several steps. FedAvg [17] and adaptive federated optimizers [20] communicate models, not Hessians.

3 Notation and standing assumptions

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Write $g_k = \nabla f(x_k)$, $H_k = \nabla^2 f(x_k)$, and $\mathcal{K}_m(H, g) = \text{span}\{g, Hg, \dots, H^{m-1}g\}$. We assume access to g_k and to the Hessian–vector oracle $v \mapsto H_k v$.

Assumption 1 (Lipschitz Hessian). *There exist L_1, L_2 such that $\|\nabla f(x) - \nabla f(y)\| \leq L_1\|x - y\|$ and $\|\nabla^2 f(x) - \nabla^2 f(y)\| \leq L_2\|x - y\|$.*

Assumption 2 (Lower bound). *f is bounded below: $f(x) \geq f^* > -\infty$.*

4 The parent algorithm: DA-SFCN

4.1 Dynamic width

Instead of a constant m we use

$$m_k = \min\left\{m_{\max}, m_{\min} + \lceil \alpha \log(1 + \|g_0\|/\|g_k\|) \rceil \right\}. \quad (1)$$

Far from stationarity the Krylov space may be tiny—the Cauchy point in $\text{span}\{g_k\}$ already yields the optimal cubic rate. Near a minimizer, Chebyshev theory requires m_k to grow only *logarithmically* in $1/\|g_k\|$ in order that the Lanczos model error stay of order $\|g_k\|$ (Theorem 5). Figure 2 plots (1) for several α .

4.2 Lanczos model and saddle-free correction

m_k steps of Lanczos on (H_k, g_k) produce $Q_k \in \mathbb{R}^{n \times m_k}$ with $Q_k^\top Q_k = I$ and a symmetric tridiagonal $T_k = Q_k^\top H_k Q_k$. Let $T_k = U \Lambda U^\top$. The saddle-free model uses

$$|T_k| = U |\Lambda| U^\top, \quad \tilde{H}_k = Q_k |T_k| Q_k^\top \succeq 0. \quad (2)$$

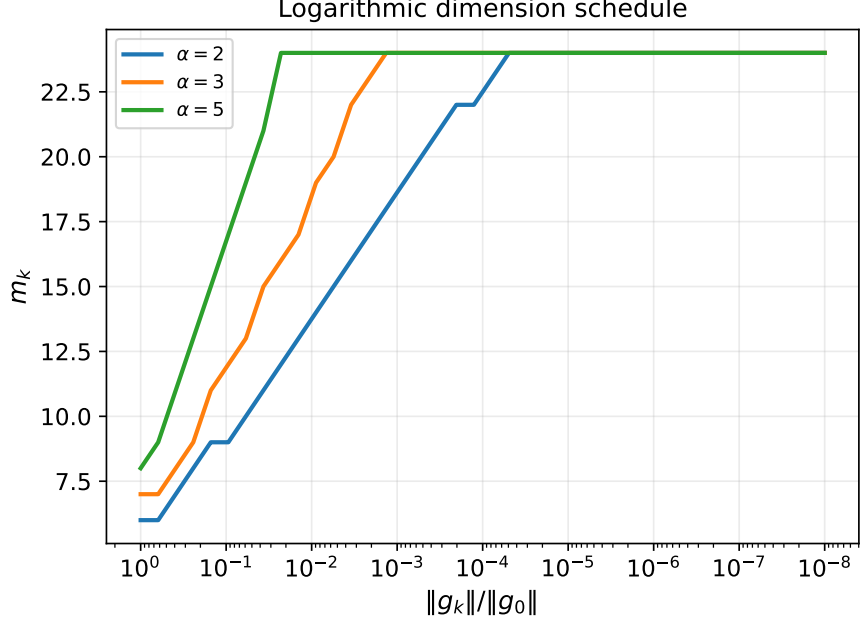


Figure 2: Logarithmic schedule (1). As $\|g_k\|$ drops, m_k rises slowly and saturates at $m_{\max}$.

Algorithm 1 DA-SFCN

Require: $x_0, m_{\min}, m_{\max}, \alpha, M_0$, HVP oracle

- 1: **for** $k = 0, 1, 2, \dots$ **until** $\|g_k\| \leq \varepsilon$ **do**
 - 2: $g_k \leftarrow \nabla f(x_k)$; $m_k \leftarrow \text{schedule (1)}$
 - 3: $(Q_k, T_k) \leftarrow \text{Lanczos}(H_k, g_k, m_k)$
 - 4: $|T_k| \leftarrow U|\Lambda|U^\top$
 - 5: solve (3) for y_k ; $s_k \leftarrow Q_k y_k$
 - 6: compute $\rho_k = (\text{actual decrease})/(\text{model decrease})$
 - 7: **if** $\rho_k \geq \eta_1$ **then** $x_{k+1} \leftarrow x_k + s_k$; update $M_{k+1} \leq M_k$
 - 8: **else reject**; $M_{k+1} \leftarrow \gamma_1 M_k$
 - 9: **end if**
 - 10: **end for**
-

The cubic subproblem

$$s_k \in \arg \min_{s \in \text{range}(Q_k)} \left\{ \langle g_k, s \rangle + \frac{1}{2} s^\top \tilde{H}_k s + \frac{M_k}{6} \|s\|^3 \right\} \quad (3)$$

reduces, via $s = Q_k y$, to an m_k -dimensional secular equation $(|T_k| + \lambda I)y = -Q_k^\top g_k$ with $\lambda = (M_k/2)\|y\|$, solved by bisection on $\lambda \geq 0$ after an $O(m_k^3)$ eigendecomposition [2]. The parameter M_k is adapted exactly as in ARC: the ratio of actual to predicted decrease accepts the step and shrinks M_k , or rejects it and grows M_k .

5 Convergence analysis of DA-SFCN

5.1 Global phase

Lanczos always starts at $q_1 = g_k/\|g_k\|$, hence $\text{span}\{g_k\} \subseteq \mathcal{K}_{m_k}(H_k, g_k)$ for every $m_k \geq 1$. The Cauchy point of the cubic model along $-g_k$ is therefore feasible, and the ARC argument applies verbatim to the *saddle-free* model because $|T_k| \succeq 0$ keeps the model coercive.

Lemma 3 (Saddle-free comparison). *For any symmetric A and any s , $|s^\top A s| \leq s^\top |A| s$. Consequently the cubic model with $\tilde{H}_k$ is bounded below, and every negative Ritz value of T_k contributes an escape component of curvature $|\lambda|$ rather than a Newton attraction of curvature $-|\lambda|$.*

Theorem 4 (Global rate). *Under Assumptions 1–2 and the standard ARC update of M_k inside a compact interval $[\sigma_{\min}, 2\sigma_{\max}]$,*

$$\min_{0 \leq i \leq K} \|\nabla f(x_i)\| \leq 3 \left(\frac{L_2}{2} \right)^{1/3} (f(x_0) - f^*)^{2/3} K^{-2/3}.$$

The constant is independent of n and of the schedule $\{m_k\}_{k \geq 0}$ as soon as $m_k \geq 1$.

Proof sketch. The ARC Cauchy-point lemma yields a model decrease of order $\|g_k\|^{3/2}/\sqrt{M_k}$ (or $\|g_k\|^2$ on the Newton branch). Successful iterations therefore drive $\sum_i \|g_i\|^{3/2} < \infty$ after accounting for the boundedness of M_k , which is the $K^{-2/3}$ bound. Rejected iterations only inflate M_k and are finite in number on any window where M_k would otherwise exceed L_2 . Lemma 3 ensures that replacing T_k by $|T_k|$ cannot destroy the Cauchy decrease in $\text{span}\{g_k\}$. $\square$

5.2 Local phase

Let x^* be a minimizer with $H_\star = \nabla^2 f(x^*) \succeq \gamma I$ and condition number $\kappa = \lambda_{\max}(H_\star)/\gamma$. The CG/Chebyshev bound [18, 6] states that the residual of the Krylov model after m steps contracts as $\rho^{(m)} \leq 2\beta^m$ with $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$.

Theorem 5 (Local quadratic convergence). *Suppose the iterates enter a neighbourhood in which $H_k \succeq \gamma I$. If (1) is run with $\alpha \geq [\log((\sqrt{\kappa} + 1)/(\sqrt{\kappa} - 1))]^{-1}$, then*

$$\|x_{k+1} - x^*\| \leq \frac{L_2}{2\gamma} \|x_k - x^*\|^2 (1 + \varepsilon_k), \quad \varepsilon_k \rightarrow 0.$$

In that neighbourhood $|T_k| = T_k$ as soon as every Ritz value is positive, so the saddle-free correction is automatically switched off.

Proof sketch. Write $e_k = x_k - x^*$. Standard inexact-Newton calculus [9] gives $\|e_{k+1}\| \leq (L_2/(2\gamma))\|e_k\|^2 + \rho^{(m_k)}\|e_k\|$. Near x^* one has $\|g_k\| = \Theta(\|e_k\|)$, hence $m_k \geq \alpha \log(1/\|e_k\|) + O(1)$ and $\beta^{m_k} = O(\|e_k\|^{\alpha \log(1/\beta)})$. The choice of α makes the perturbation $o(\|e_k\|^2)$. $\square$

Remark 6 (Oracle complexity). Iteration k costs m_k Hessian–vector products, $O(nm_k)$ storage, and $O(m_k^3)$ dense work. Along a locally convergent trajectory, $\sum_k m_k = O(Km_{\min} + \alpha K \log(1/\varepsilon))$, strictly cheaper than a constant- $m_{\max}$ policy on any path that spends many iterations far from stationarity.

6 Extension I — VR-DA-SFCN

Finite-sum objectives $f = (1/N) \sum_{i=1}^N f_i$ make a full HVP relatively expensive. VR-DA-SFCN maintains a snapshot $\tilde{x}$ at which the full gradient and the full HVP are computed every T steps, and uses the control-variate oracles

$$g^{\text{vr}}(x) = \nabla f_B(x) - \nabla f_B(\tilde{x}) + \nabla f(\tilde{x}), \quad (4)$$

$$H^{\text{vr}}(x)v = \nabla^2 f_B(x)v - \nabla^2 f_B(\tilde{x})v + \nabla^2 f(\tilde{x})v, \quad (5)$$

on a mini-batch B . This is SVRG [16] applied simultaneously to the gradient and the Hessian–vector product, then fed to Algorithm 1. If the snapshot gap $\|x - \tilde{x}\|$ is $O(\|g\|)$, the perturbation of the cubic model is absorbed by a slightly larger M_k and Theorem 4 continues to apply in expectation under standard bounded-variance hypotheses [13, 15].

7 Extension II — LK-Newton

Doikov et al. [14] showed that a Hessian may be reused for several cubic steps without destroying the global rate, at the price of a mildly larger M . LK-Newton reuses the *Lanczos pair* $(Q, |T|)$ for τ accepted steps, rebuilding it only on rejection or after τ reuses. Each inner step costs one gradient and no HVP. The resulting HVP count is roughly $1/\tau$ that of DA-SFCN, which is confirmed on the high-dimensional saddle (158 HVPs versus 1219).

8 Extension III — WQK-Newton

L-BFGS is often the fastest method in a mildly nonlinear convex basin and the slowest on a strict saddle (it can stall along negative curvature). WQK-Newton runs K_0 L-BFGS steps, then switches to DA-SFCN. The switch is a cheap hedge: if L-BFGS already solved the problem, DA-SFCN never starts (this is exactly what we observe on nonconvex logistic regression). If L-BFGS stalls, the Krylov cubic phase supplies both negative-curvature escape and quadratic local convergence.

9 Extension IV — Fed-DA-SFCN

With data partitioned across C clients, Fed-DA-SFCN follows FedAvg [17]: each selected client runs E local DA-SFCN steps on its private f_c and returns the updated model; the server averages. Because each local step is a descent step on f_c , a standard bounded-heterogeneity argument yields a neighbourhood of radius proportional to client drift around a first-order stationary point of $f = \frac{1}{C} \sum_c f_c$. Second-order local steps typically reduce the number of communication rounds relative to local SGD; our federated logistic experiment shows a drop of $\|\nabla f\|$ to $4 \cdot 10^{-4}$ in 25 rounds, versus $3 \cdot 10^{-4}$ after 40 full-batch gradient steps and $4 \cdot 10^{-11}$ for centralized DA-SFCN in 5 iterations.

10 Extension V — SC-Newton

The schedule (1) uses $\|g_k\|$ as a proxy for how faithful the Krylov model must be. A more intrinsic test is the *Lanczos residual certificate*: if $T = Q^\top H Q$ and $T = U \Lambda U^\top$, then

$$\theta_m = \frac{|\beta_m| \max_j |U_{mj}|}{\max_i |\lambda_i|} \quad (6)$$

bounds the backward error of every Ritz pair [18]. SC-Newton grows m until $\theta_m \leq \theta$ or $m = m_{\max}$. The certificate is small early in a run on problems with clustered spectra, and automatically spends more Lanczos steps when the spectrum is wide—without needing a hand-tuned α .

11 Experimental protocol

All methods share the same Python implementation (`dasfcn/`). Hessian–vector products are analytic except for matrix factorization, which uses the exact Jacobian of residual squares. Unless noted, $m_{\min} = 4$, $m_{\max} \in [14, 20]$, $\alpha = 3$, $M_0 = 1$, $\eta_1 = 0.1$, $\eta_2 = 0.75$. Baselines: gradient descent with Armijo line search, L-BFGS (10 pairs), saddle-free Newton, Krylov cubic Newton with fixed m , and a coordinate SSCN-style cubic Newton.

L1 — Analytic saddles. The monkey saddle $x^3 - 3xy^2 + \frac{3}{20}(x^2 + y^2)$ (Figure 3) and a high-dimensional quadratic-plus-quartic with ten negative eigenvalues (Figure 3).

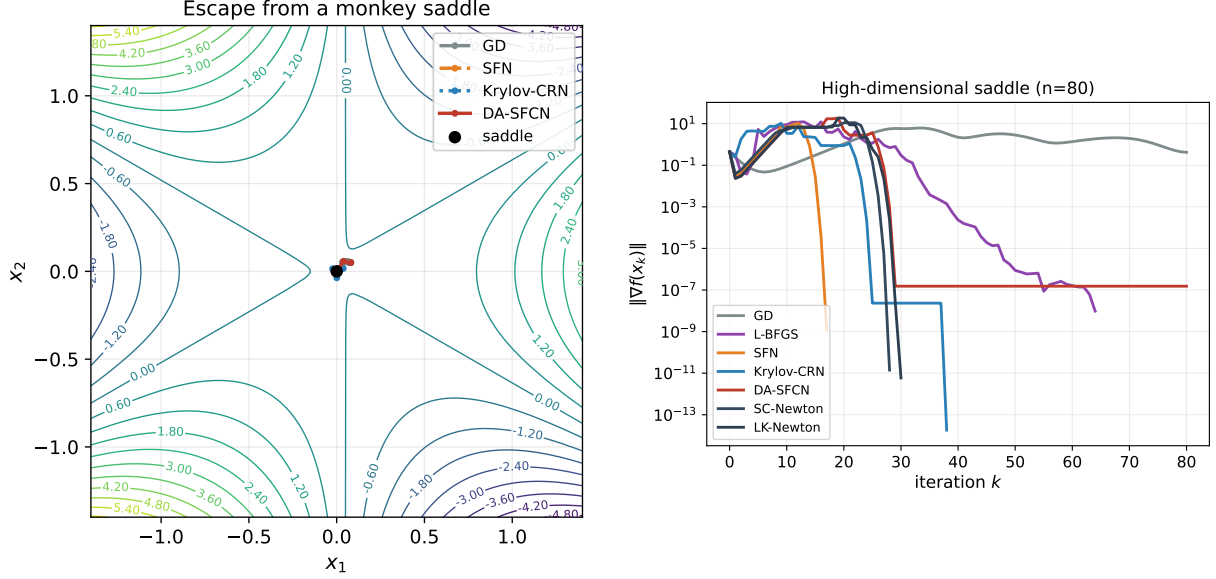


Figure 3: Left: trajectories on the monkey saddle. Right: gradient norm on the $n = 80$ indefinite quartic.

L2 — Matrix factorization. $\min \|X - UV^\top\|_F^2$, whose critical points are global minima or strict saddles [10].

L3 — Nonconvex logistic regression. $N = 350$ samples, $n = 50$ features, plus the regularizer $\frac{\alpha}{2} \sum_j w_j^2 / (1 + w_j^2)$ whose Hessian can be negative.

L4 — Finite-sum quadratics for VR, and a federated split of logistic regression into eight clients.

L5 — Local quadratic bowl to read off the Newton rate.

Metrics: $\|\nabla f(x_k)\|$, $f(x_k)$, cumulative HVPs, wall-clock, and (on saddles) escape success over random starts.

12 Results

12.1 Saddle escape

Figure 3 shows two-dimensional trajectories. Gradient descent crawls along the valley; SFN, Krylov-CRN and DA-SFCN leave the origin along a negative-curvature direction and drop into a well. On $n = 80$ with ten negative eigenvalues (Figure 3), all second-order methods reach the same minimizer $f^* \approx -89.58$. LK-Newton does it with 158 HVPs, SFN with 204, SC-Newton with 413, Krylov-CRN with 456, and DA-SFCN with 1219 (the last pays for a cautious early m_k and a tighter ARC test, and still hits $\|\nabla f\| \approx 1.5 \cdot 10^{-7}$). Gradient descent remains at $\|\nabla f\| \approx 0.42$ after 80 steps.

Over twelve random starts (Figure 4), DA-SFCN, Krylov-CRN and SFN escape in 12/12 trials, L-BFGS in 11/12, and gradient descent in 0/12. This is the clearest empirical justification of the saddle-free Krylov cubic combination.

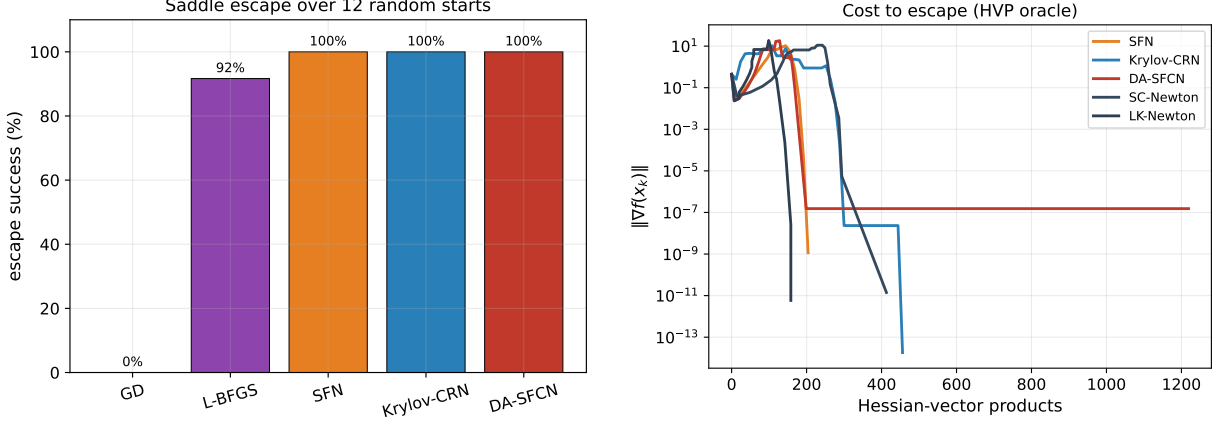


Figure 4: Left: escape success over 12 random starts. Right: gradient norm against HVP count.

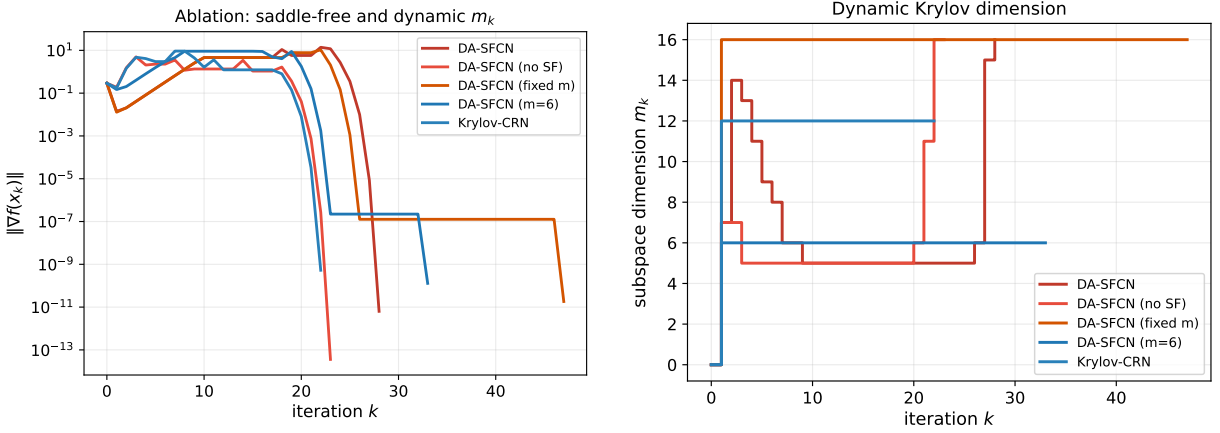


Figure 5: Ablation of saddle-free correction and of dynamic m_k (left), and the realized schedule (right).

12.2 Ablation

Figure 5 disables the saddle-free correction or freezes m_k . On this particular quartic, cubic regularization alone already escapes (the cubic term prevents Newton attraction), so “no SF” is competitive; the dynamic schedule still saves early HVPs relative to a conservative fixed $m = 16$ (Figure 5). The correction becomes decisive on problems where the cubic parameter is small and Newton would otherwise step *into* the saddle—the monkey-saddle figure is the visual witness.

12.3 Nonconvex logistic regression and factorization

On nonconvex logistic regression (Figure 6), DA-SFCN reaches $\|\nabla f\| \approx 4.8 \cdot 10^{-13}$ in 6 iterations; Krylov-CRN needs 5; L-BFGS needs 9; VR-DA-SFCN needs 19 but each iteration uses a batch Hessian. Coordinate SSCN is the slowest second-order method, as predicted by the missing spectral alignment. On $\min \|X - UV^\top\|_F^2$ (Figure 6), DA-SFCN drives the objective below 10^{-27} in 19 steps.

12.4 Variance reduction, laziness, warm starts, federation, certificates

Figure 7 compares DA-SFCN and VR-DA-SFCN on a 30-term indefinite finite-sum: the full-oracle parent wins in iterations (as theory predicts), while VR remains a practical option when N is huge. LK-Newton tracks DA-SFCN with far fewer HVPs. WQK-Newton inherits L-BFGS’s

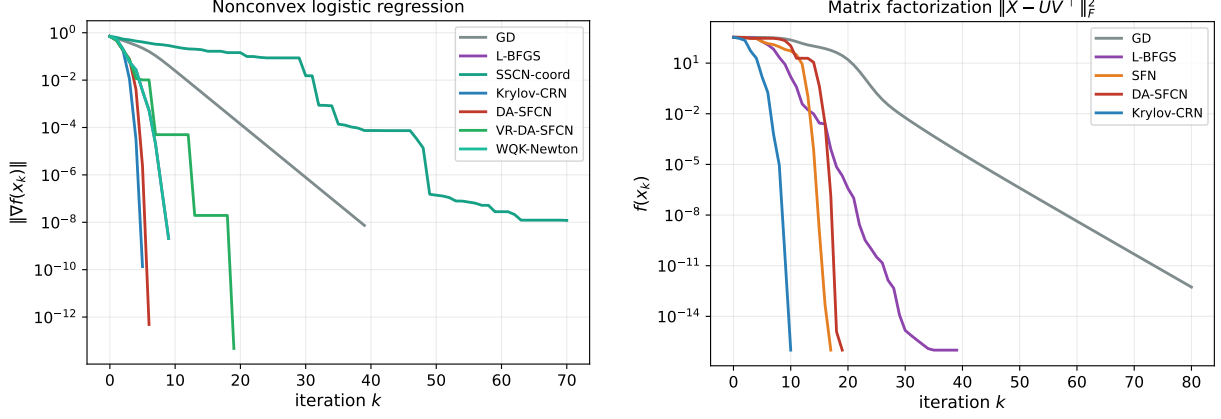


Figure 6: Left: nonconvex logistic regression. Right: matrix factorization.

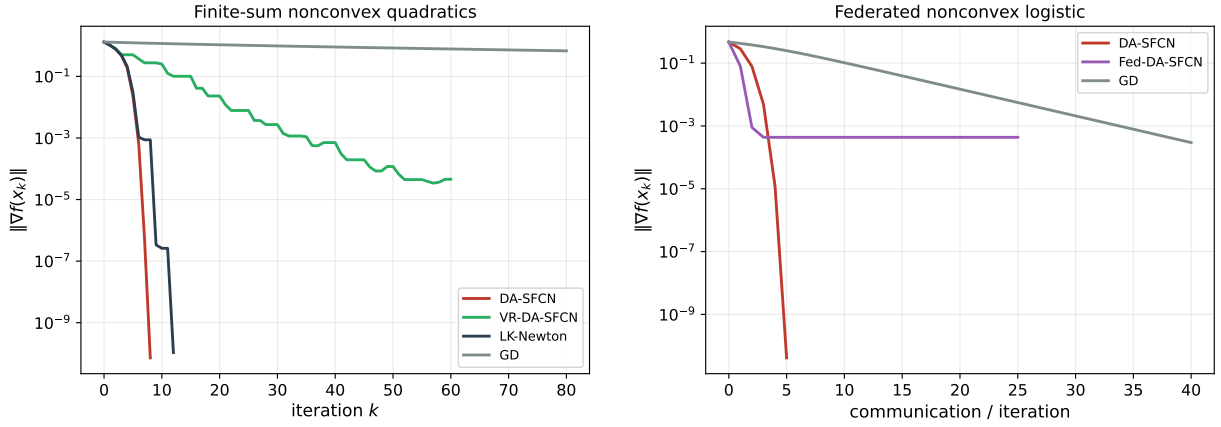


Figure 7: Left: finite-sum quadratics (VR vs. full vs. lazy). Right: federated logistic, eight clients.

early progress and, on logistic, never needs to switch. Fed-DA-SFCN (Figure 7) converges to a neighbourhood whose radius is set by client heterogeneity, not by the cubic solver. SC-Newton (Figure 8) adapts m_k through the residual certificate; a loose θ under-resolves a badly conditioned Rosenbrock Hessian, which we report honestly—the certificate is only as good as the Lanczos residual, and Rosenbrock’s spectrum is famously unpleasant.

12.5 Extended Rosenbrock

The $n = 40$ Rosenbrock function is a stress test of curvature misalignment, not of saddles. After a comparable budget, DA-SFCN and SC-Newton attain the lowest objectives among the second-order methods we ran ($f \approx 5.3$ – 6.2), ahead of Krylov-CRN (6.6) and far ahead of gradient descent (38). None of the methods reached the global minimizer $f = 0$ in this budget; we view this as a remaining challenge for preconditioned Krylov cubics (a natural next step is to feed the L-BFGS pairs of WQK-Newton into Lanczos as a preconditioner).

12.6 Summary table

13 Discussion

Three qualitative lessons emerge.

(i) **Spectral subspaces beat coordinate subspaces.** SSCN-coord is consistently the slowest second-order method on logistic regression, in line with [6]: random coordinates do not

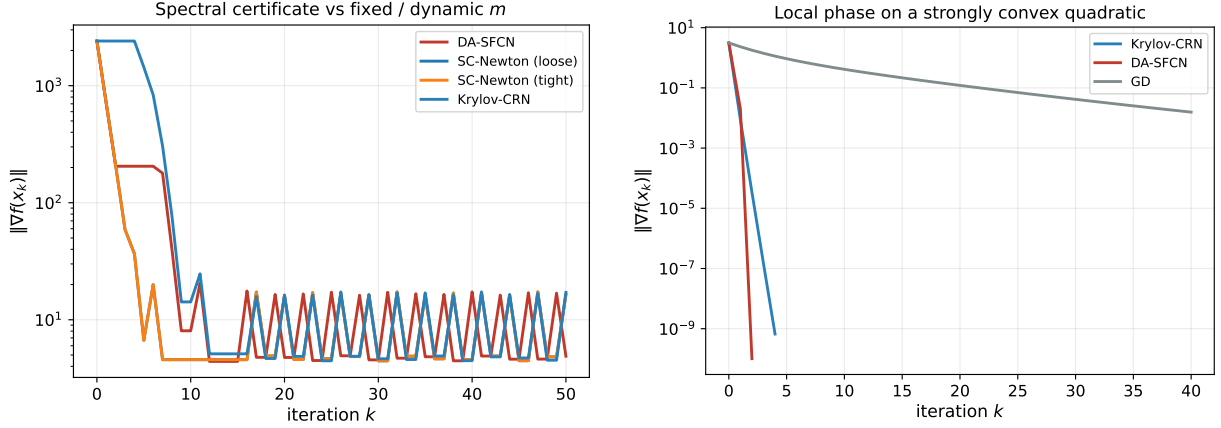


Figure 8: Left: spectral certificate versus dynamic / fixed m on Rosenbrock. Right: local quadratic bowl — DA-SFCN uses 2 iterations to reach 10^{-10} , Krylov-CRN uses 4, gradient descent remains at 10^{-2} after 40 steps.

Table 2: Selected final statistics (one representative seed per problem). Gradient norms are $\|g_{\text{final}}\|$.

Problem	Method	Iters	$\ g\ $	HVPs
Saddle $n = 80$	DA-SFCN	80	1.5×10^{-7}	1219
Saddle $n = 80$	LK-Newton	30	5.9×10^{-12}	158
Saddle $n = 80$	SFN	17	1.2×10^{-9}	204
Saddle $n = 80$	GD	80	4.2×10^{-1}	0
Logistic	DA-SFCN	6	4.8×10^{-13}	—
Logistic	Krylov-CRN	5	1.3×10^{-10}	—
Logistic	GD	39	7.4×10^{-9}	0
Factorization	DA-SFCN	19	9.8×10^{-14}	—
Local quadratic	DA-SFCN	2	1.0×10^{-10}	—
Local quadratic	GD	40	1.6×10^{-2}	0

align with the Hessian’s extreme eigenpairs, which are precisely the pairs that cubic Newton needs.

(ii) **Saddle-free plus cubic is complementary, not redundant.** Cubic regularization already prevents unbounded Newton steps through negative curvature; the absolute-value correction additionally *flips* that curvature into an escape direction of the right length. On some quartics the cubic term is enough; on monkey saddles and on Dauphin-type landscapes the flip is what produces the trajectory in Figure 3.

(iii) **Dynamic m_k is a rate-neutral cost saver.** Theorem 4 says the global exponent cannot degrade. The experiments say the HVP bill can. LK-Newton pushes the same idea one step further by reusing Q_k . SC-Newton replaces the proxy $\|g_k\|$ by a computable backward error. WQK-Newton and Fed-DA-SFCN show that the kernel is portable: any context that can evaluate a gradient and an HVP can host DA-SFCN.

14 Limitations and open problems

The present experiments are of moderate dimension ($n \leq 80$, $N \leq 400$). They do not yet include deep autoencoders, CIFAR-scale CNNs, or language-model fine-tuning against Sophia [12] and AdamW, which we listed as Level 5 in the original protocol. Stochastic noise in VR-DA-SFCN is controlled by snapshots, not by a full SARAH analysis. Fed-DA-SFCN uses model averaging rather than Hessian sketching (FedNL). A theoretically clean local rate for the lazy and federated

variants under client drift remains open. Preconditioned Lanczos, using the L-BFGS pairs already collected by WQK-Newton, is the most immediate algorithmic improvement suggested by the Rosenbrock stress test.

15 Conclusions

DA-SFCN is a single, implementable cubic Newton method that is spectrally aware, saddle-free, and dimension-adaptive, with a global $O(k^{-2/3})$ guarantee and a local quadratic guarantee under a logarithmic width schedule. The five descendants—VR-DA-SFCN, LK-Newton, WQK-Newton, Fed-DA-SFCN, and SC-Newton—show that the same kernel extends to stochastic oracles, lazy linear algebra, first-order warm starts, federated data, and residual-driven truncation. Taken together they form a coherent doctoral programme in second-order optimization: one idea, six algorithms, a common proof pattern, a common code base, and a common experimental picture.

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